Comparing RL Algorithms, Rewards, and Policy Architectures for Token-Level Prompt Compression

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Introduction & Problem Statement

LLM inference cost scales with prompt length. We formulate **prompt compression** as token-level RL: agent emits binary keep/drop per token, compressed prompt fed to frozen LLM, reward from task performance.

Setup: 6 RL algorithms (REINFORCE, MCTS, A2C, GRPO, PPO, contextual bandit) $\times$ 4 rewards (sparse, KL-dense, hybrid, learned) $\times$ 4 architectures (DistilBERT/DistilRoBERTa, TinyLlama, custom Transformer) on SQuAD 2.0[7].

Notation: $\mathcal{M}$ = frozen LLM, $\mathbf{x} = (x_1, \ldots, x_N)$ = tokens, $\tilde{\mathbf{x}} \subseteq \mathbf{x}$ = compressed prompt, $\rho = |\tilde{\mathbf{x}}|/N$. Overlapping chunks (128 tokens, overlap 16).

Prior Work: PCRL[1] (RL + gated reward + REINFORCE for token compression), TACO[2] (contrastive-learning compressor, no algorithm/reward ablation). We systematically compare across all three axes.

Challenge: Sparse reward at aggressive compression; entropy regularization critical to avoid keep-all or drop-all collapse.

Datasets & Baselines

SQuAD 2.0[7]: 100K+ extractive QA pairs with 50K unanswerable questions. Models must extract an answer span from the context or abstain. Metrics: token-level F1 and exact match (EM). The frozen LLM receives the full compressed prompt (question + context) and generates an answer.

Baselines: (1) Random drop (keep each token with probability = target ratio), (2) Selective Context[8] (retain by self-information), (3) LLMLingua-2[9] (BERT token classifier; 512-token limit).

LLMLingua-2 best at low ρ; Selective Context can underperform random.

Limitations & Future Work

Entropy collapse: PPO collapsed 3/3 seeds; A2C $\to \rho$=0 or 1. Learned value functions vulnerable to sparse reward; only value-free methods (REINFORCE, bandit) stable. Fix: adaptive entropy floors, KL constraints.

Compression ceiling: Best $\rho \approx$0.74–0.80, far above LLMLingua-2 (ρ=0.30). Curriculum ρ^*: 0.9 $\to$ 0.5 could help.

Model capacity bottleneck: PCRL's gated reward uses τ=0.9 faithfulness threshold; we dropped to τ=0.6 because Llama-3.2-1B is too sensitive to token removal. Even mild compression degrades output quality below 0.9. A larger target LLM (e.g. 8B+) would likely tolerate more aggressive dropping, raising the faithfulness floor and enabling the stricter threshold.

Single domain: SQuAD 2.0 (English extractive QA) only. Transfer across tasks (summarization, code), languages, and LLM families unstudied. Hard token-dropping may also lose to soft compression at aggressive ρ.

References

[1] H. Jung and D. Kim. PCRL: Prompt Compression with Reinforcement Learning. *IEEE Access*, 2024.
[2] J. Li, S. Chen, W. Liu, et al. TACO: Token-Aware Prompt Compression via Contrastive Learning. *EMNLP*, 2024.
[3] C.-Y. Lin. ROUGE: A Package for Automatic Evaluation of Summaries. *ACL Workshop on Text Summarization*, 2004.
[4] R. J. Williams. Simple Statistical Gradient-Following Algorithms for Connectionist Reinforcement Learning. *Machine Learning*, 8(3–4):229–256, 1992.
[5] S. J. Rennie, E. Marcheret, Y. Mroueh, J. Ross, and V. Goel. Self-Critical Sequence Training for Image Captioning. *CVPR*, 2017.
[6] D. Silver, J. Schrittwieser, K. Simonyan, et al. Mastering the Game of Go Without Human Knowledge. *Nature*, 550(7676):354–359, 2017.
[7] P. Rajpurkar, R. Jia, and P. Liang. Know What You Don't Know: Unanswerable Questions for SQuAD. *ACL*, 2018.
[8] Y. Li, B. Dong, F. Guerin, and C. Lin. Compressing Context to Enhance Inference Efficiency of Large Language Models. *EMNLP*, 2023.
[9] Z. Pan, Q. Wu, H. Jiang, et al. LLMLingua-2: Data Distillation for Efficient and Faithful Task-Agnostic Prompt Compression. *ACL*, 2024.
[10] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal Policy Optimization Algorithms. *arXiv:1707.06347*, 2017.
[11] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel. High-Dimensional Continuous Control Using Generalized Advantage Estimation. *arXiv:1506.02438*, 2015.

A2C vs GRPO

Compare: A2C and GRPO on SQuAD 2.0 with hybrid reward. 250 episodes, 16-episode batches.

A2C: Converges to ρ=0 or 1 depending on seed; reward collapses or shows high variance, suggesting reward hacking.

GRPO: Both Custom and DistilBERT display decreasing ρ. DistilBERT shows increasing reward with high variance; Custom Transformer fails to learn.

Hybrid Reward:

$$R_{\text{episode}} = \sum_{t \in \text{kept}} r_t + r_{\text{terminal}}$$

$$r_t = -\lambda_{\text{KL}} \, \mathrm{KL}\Big(p_\theta(\cdot \mid x_{<t}) \,\Big\|\, p_{\text{ref}}(\cdot \mid x_{<t})\Big)$$

$$r_{\text{terminal}} = F1(y, y^*)\,(1-\rho), \qquad \rho = \frac{|\text{kept tokens}|}{|\text{total tokens}|}$$

Configuration	Reward	Entropy	Compression Ratio
Vanilla A2C Custom	0	0.005	1
GRPO Custom	−1.579	0.664	0.358
GRPO DistilBERT	−0.189	0.075	0.019

Table 1. Eval at 250 ep. A2C collapses to ρ=0 or 1; GRPO DistilBERT learns but compresses too aggressively.

PPO vs Contextual Bandit

Compare: PPO[10] and contextual bandit on SQuAD 2.0 with sparse additive reward. Same frozen encoder/LLM as REINFORCE/MCTS. 300 episodes, 3 seeds, 8-episode batches.

Reward: Ground-truth F1 is tiny ($\sim$0.01) at moderate compression; scale and add ratio penalty at ρ^*=0.5:

$$r = \tau \cdot \mathrm{F1}(\hat{y}, y^*) - \lambda\,|\rho - \rho^*| + b, \qquad \tau=20,\ \lambda=1,\ b=0.15$$

PPO[10]

Clip ε=0.2, GAE[11] λ=0.95, entropy coeff 0.1, 2 epochs/batch. Terminal reward distributed equally across kept tokens.

Contextual Bandit

Each token is an independent arm; ε-greedy (0.4$\to$0.01, decay 200 steps), entropy reg. (β=0.1). No value network, no temporal credit assignment.

Configuration	ρ	Reward	F1	Entropy
Bandit (mean ± std)	0.509	0.302	0.010	0.692
PPO (2/3 seeds)	0.078	−0.273	0.0	0.000
PPO (1/3 seeds)	1.000	−0.350	0.0	0.000

Table 2. PPO: entropy collapse in 3/3 seeds within 80–160 episodes. Two seeds drop-all ($\rho \approx$0.07); one keeps-all (ρ=1.0). Bandit: Stable; entropy $\approx \ln 2$, ratio converges to ρ^*=0.5 via penalty alone.

Conclusion

Reward design is most critical. Faithfulness ROUGE-L (5–10$\times$ stronger than ground-truth F1) enabled REINFORCE/MCTS; ground-truth F1 $\to$ PPO collapse, bandit near-zero F1. Hybrid KL-dense: GRPO learns, A2C still collapses.

Simplicity wins. Frozen encoder + lightweight head most stable across all algorithms; fine-tuning or heavier networks $\to$ entropy collapse. Bandit matched PPO (no temporal credit); REINFORCE > MCTS in peak F1; GRPO > A2C.

REINFORCE vs MCTS

Compare: (1) {REINFORCE, MCTS} $\times$ {gated[1], continuous}
(2) {MCTS} $\times$ { soft-gated, harmonic reward, self-attention head variant}

Constants: Frozen DistilRoBERTa-base encoder (82M params) with a lightweight trainable head with frozen Llama-3.2-1B-Instruct (4 bit) as the target LLM.

Metric: Faithfulness computed as ROUGE-L[3] which scores how well the compressed-output response preserves the content and ordering of the original-output response via longest common subsequence overlap.

Reward Functions, $f = \text{ROUGE-L}(\mathcal{M}(\tilde{\mathbf{x}}), \mathcal{M}(\mathbf{x}))$:

Smooth	Gated[1]	Soft-gated	Harmonic
$f(1-\rho)$	$(1-\rho)\,\mathbf{1}\{f \ge \tau\} - p\,\mathbf{1}\{f < \tau\}$	$\sigma(k(f-\tau))(1-\rho)$	$2f(1-\rho)(f+(1-\rho))^{-1}$

Classification head: (1) MLP ($\sim$197K params; score tokens independently). (2) Self-attention ($\sim$460K params; score importance via 4-head attention).

Both 0-init final layer with bias $[0, 1]$ to yield initial keep probability $\frac{e}{1+e} \approx 73\%$.

REINFORCE[4]

$k = 8$ stochastic rollouts plus 1 greedy rollout per prompt. Hybrid SCST/EMA baseline[5] to compare sampled against greedy rollouts.

$$\mathcal{L} = -\sum_i \log \pi_\theta(a_i \mid \mathbf{x})\,\hat{A} - \beta\bar{\mathcal{H}}$$

$$\hat{A} = R - \tfrac{1}{2}R_{\text{greedy}} - \tfrac{1}{2}\bar{R}_{\text{EMA}}$$

MCTS[6]

Search tree with one chunk per depth. Uses a heuristic value estimate (no learned value network) and policy priors.

$$a^* = \arg\max_a \left[Q(a) + c_{\text{puct}} P(a) \frac{\sqrt{N_{\text{par}}}}{1 + N_a} \right]$$

$$V(s) = \bar{f}_{\text{EMA}} \cdot \max(0,\ 1 - \rho_s)$$

Shared Hyperparameters: lr $= 3 \times 10^{-5}$, $k = 8$ samples, 128 episodes/step, max grad norm 5.0, $\beta = 0.001$, $\tau = 0.6$.

	RF Gate	MCTS Gate·Attn	MCTS Harmonic	MCTS Soft-gate	MCTS Cont	RF Cont	MCTS Gate	LLMLingua-2	Random
F1 ↑	**.221**	.214	.180	.174	.168	.151	.129	.037	.018
ρ	.803	.737	.766	.772	.712	.707	.748	.300	.700

REINFORCE vs MCTS: Reward Stability and Exploration

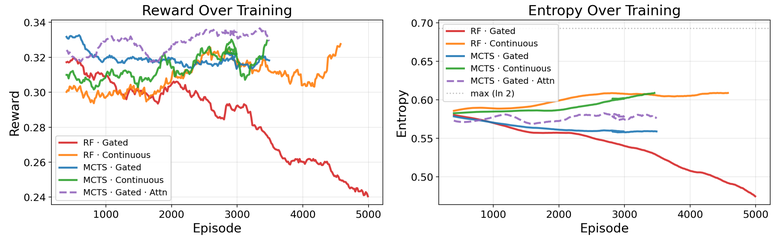

Figure 1. REINFORCE achieves the highest peak F1 (0.221) but at conservative compression (ρ=0.80). MCTS with a self-attention head nearly matches (0.214) at ρ=0.74, yielding a better compression–quality tradeoff. Gated reward peaks higher but exhibits reward drift under extended training. Continuous reward is more stable but peaks $\sim$30% lower. Self-attention boosts MCTS-gated F1 from 0.129 to 0.214 (+66%), showing that cross-token importance scoring matters for compression.